

PH.D. QUALIFYING EXAM 2024: PROBABILITY

Rules: This is a four hour exam consisting of two Parts - Part I and Part II, both of which should be answered. Part I contains **three independent problems**. Part II consists of **ten questions on one problem**. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “information sheet” for EACH OF THE TWO COURSES (Stat 592 and Stat 654) on which this exam is based. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly in the space provided after each problem, **preferably using pens**, leaving an impression deep enough for the papers to be scanned. Please use only one-side of the pages to write your answers and insert blank pages if you need more space to write an answer. Good luck!

PART I**PROBLEM 1**

Suppose X has a uniform distribution on $[0, 1]$, and let N represent the first decimal place for which the digit is non-zero. For example, if $X = 0.00532$, then $N = 3$. Find the mean and variance of N . Your final answer should be given as two simple numerical values.

PROBLEM 2

X has a Geometric(p), $0 < p < 1$, distribution if it has a probability mass function of the form $f(x; p) = (1 - p)^{x-1}p$ for $x \in \{1, 2, \dots\}$. Suppose X_1 and X_2 are independent, with X_j having a Geometric(p_j) distribution for $j = 1, 2$. Define $Y = X_1 + X_2$.

- a) Find the moment generating function of Y if it exists, or otherwise verify that it does not exist.

- b) Find the probability mass function of Y . Your answer should be given in a simple closed form.

- c) Prove or disprove: For each fixed $y \in \{1, 2, \dots\}$, the probability mass function of Y , say $f(y; p_1, p_2)$ is continuous as $p_1 \rightarrow p_2$.

PROBLEM 3

Suppose X has density of the form $f(x) = c \frac{\log x}{x^2}$ for $x > 1$.

- a) Find the value of the normalizing constant c .

- b) Find $\mathbb{E}[X^k]$ for $k = 1, 2, \dots$ if it exists, or otherwise verify it does not exist.

- c) Find the moment generating function of X if it exists, or otherwise verify it does not exist.

- d) If X_1, X_2 are independent copies of X , find the density of $Y = X_1 X_2$.

PART II

Let (V, E) be an undirected graph with a finite set $V = \{1, \dots, n\}$ of vertices. Two vertices v, v' are connected if and only if $\{v, v'\} \in E$. We write $v \sim v'$ if the two vertices are connected.

Let $\mathcal{X} = \{-1, 1\}^n$ be the state space. This corresponds to a configuration of spins, either $+1$ or -1 , on each vertex. Define a function $H : \mathcal{X} \rightarrow \mathbb{R}$ by

$$H(x) = \sum_{\{v, v'\} \in E} 1\{x(v) \neq x(v')\} = \frac{1}{2} \sum_{\{v, v'\} \in E} (1 - x(v)x(v'))$$

where $1\{x(v) \neq x(v')\}$ an indicator function value in $\{0, 1\}$. Let

$$\pi(x) = \frac{\exp(-H(x))}{\sum_{y \in \mathcal{X}} \exp(-H(y))}.$$

Consider the following Markov Chain update: Given a current state x_t , sample the next state $X_{t+1} = f_{t+1}(x_t)$ where the random function $f_{t+1} : \mathcal{X} \rightarrow \mathcal{X}$ is defined as follows:

Algorithm 1 Random function f_{t+1}

Input: $x_t \in \mathcal{X}$

Sample $v_{t+1} \in V$ uniformly at random,

Sample $S_{t+1} \in \{-1, 1\}$ uniformly at random independently of v_{t+1} ,

Sample $U_{t+1} \sim \text{Unif}[0, 1]$ independently of (v_{t+1}, S_{t+1}) .

Propose a move y defined as $y(v_{t+1}) = S_{t+1}$ and $y(w) = x_t(w)$ for all $w \neq v_{t+1}$.

if $U_{t+1} \leq \min\{1, \frac{\exp(-H(y))}{\exp(-H(x_t))}\}$ **then**

Accept the move and set $f_{t+1}(x_t) = y$

else

Reject the move and set $f_{t+1}(x_t) = x_t$.

end if

For two states x, z , define the transition matrix as $P(x, z) = \mathbb{P}(f_{t+1}(x) = z)$.

- (1) Compute $P(x, z)$ for any two states $x, z \in \mathcal{X}$ such that $P(x, z) > 0$ and $z \neq x$. You need not compute $P(x, x)$.

- (2) Prove that π is a stationary distribution for this chain.

- (3) Inside the algorithm that defines f_{t+1} , denote by $N_{t+1} \subset V$ the random subset containing the neighbors of v_{t+1} and denote by $y \in \mathcal{X}$ be the proposed move (as in the algorithm's definition). Show that

$$\frac{\exp(-H(y))}{\exp(-H(x_t))} = \exp\left(\frac{1}{2} \sum_{w \in N_{t+1}} \left(S_{t+1} - x_t(v_{t+1})\right) x_t(w)\right).$$

- (4) Write $x \preceq y$ if $x(v) \leq y(v)$ for all vertices $v \in V$. Prove rigorously that if $x \preceq z$ then $\mathbb{P}(f_{t+1}(x) \preceq f_{t+1}(z)) = 1$. You may handle the cases $S_{t+1} = +1$ and $S_{t+1} = -1$ separately.

- (5) Let $\underline{x} = (-1, -1, \dots, -1)$ contain only the value -1 , n times, and let $\bar{x} = (1, 1, \dots, 1)$ contain only the value $+1$, n times. Prove rigorously that $F = f_t \circ f_{t-1} \circ \dots \circ f_1$ is a constant function if and only if $F(\underline{x}) = F(\bar{x})$.

Let $T = \min\{t \geq 1 : f_t \circ f_{t-1} \circ \dots \circ f_1(\underline{x}) = f_t \circ f_{t-1} \circ \dots \circ f_1(\bar{x})\}$. Our goal in the next questions is to show that $\mathbb{E}[T]$ is not much larger than the mixing time defined as usual by

$$d(t) = \max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|_{TV}, \quad t_{\text{mix}} = \min\{t \geq 0 : d(t) \leq 1/4\}.$$

We also define, as usual, $\bar{d}(t) = \max_{x, y \in \mathcal{X}} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{TV}$.

- (6) For any state $z \in \mathcal{X}$, let $h(z)$ be the maximal number k of distinct states $x_1, \dots, x_k$ such that $x_1 \preceq x_2 \preceq \dots \preceq x_k = z$. Prove that if $x_t \preceq y_t$ and $x_t \neq y_t$ then $h(x_t) + 1 \leq h(y_t)$.

- (7) Let $X_0 = x$ and $X_t = f_t \circ \cdots \circ f_1(x)$, and similarly $Y_0 = \bar{x}$ and $Y_t = f_t \circ \cdots \circ f_1(\bar{x})$, so that $\{T > t\} = \{X_t \neq Y_t\}$. Prove rigorously that

$$\mathbb{P}(T > t) \leq \mathbb{E}[h(Y_t) - h(X_t)].$$

- (8) Prove rigorously that $0 \leq h(z) \leq |V|$ for any $z \in \mathcal{X}$, and that $\mathbb{E}[h(Y_t) - h(X_t)] \leq |V|\bar{d}(t)$.

- (9) Assume that $|V| = 2^L$ for some integer $L > 1$. Let $k \geq 0$ be another integer. Prove rigorously that

$$\mathbb{P}\left(T > t_{\text{mix}}(k + L)\right) \leq \frac{1}{2^k}.$$

Carefully state any property of $d(t)$, $\bar{d}(t)$ or t_{mix} that are used for the proof.

- (10) Prove that $\mathbb{E}[T] \leq cLt_{\text{mix}}$ for some absolute constant $c > 0$. You may first prove as an intermediate result that any non-negative random variable U (not necessarily integer valued) with $\mathbb{P}(U > k) \leq 1/2^k$ for all integer $k \geq 0$ has finite expectation smaller than a finite constant $c > 0$. The constant $c > 0$ need not be the best possible.